Generalizing Score based Generative Model Applications

Anonymous Author(s)
Affiliation
Address
email

Abstract

It is not clear in the literature if SDE based diffusion models can be used for any distribution. In particular it is not clear if the theoretical backward equation is well posed in any case and if the standard approximation using a trained score and a gaussian initialization is sufficient for approximating any distribution. In recent literature in particular a debate around the possibility of applying diffusion models to heavytail distributions is grown. In this paper we theoretically show that the use of early stopping and the replacement of the gaussian initialization with a suitable one allows the generalization of the framework of diffusion models to any distribution. In particular we show that the backward is well posed and we generalize a KL-based theorem to any existent distribution by replacing the gaussian initialiation, that is not sufficient, with a suitable one. We test our idea on toy datasets, classical datasets and heavytail datasets, showing that this framework is efficient in the number of steps and improves the generation in the heavytail context.

1 Introduction

In recent years, generative models have emerged as a cornerstone of machine learning due to their capacity to sample high-dimensional distributions. Score-based generative models (SGMs) were introduced in their discrete form by Ho et al. [Ho et al., 2020] and in their continuous formulation by Song et al. [Song et al., 2021], building on several years of pioneering work [Sohl-Dickstein et al., 2015, Song and Ermon, 2019]. Since then, they have achieved state-of-the-art performance in image and video generation, surpassing GANs, VAEs, and Normalizing Flows [Dhariwal and Nichol, 2021, Müller-Franzes et al., 2022]. Their frameworks form the foundation of Stable Diffusion [Rombach et al., 2022] and many of the most prominent publicly available models today, including DALL·E 3 [Betker et al., 2023], Imagen 3 [Baldridge et al., 2024], and recent advanced architectures like FLUX [Black Forest Labs, 2025]. SGMs have also demonstrated competitive results in audio [Chen et al., 2021, Kong et al., 2021] and text generation [Li et al., 2022, Zou et al., 2023], positioning them as a flexible and powerful framework across diverse data modalities.

The core idea of SGMs is twofold: the data distribution is gradually corrupted by adding Gaussian noise, and a neural network is trained to reverse this process, recovering the original sample from its corrupted version. New samples are then generated by starting from standard Gaussian noise and iteratively applying the trained network to progressively denoise it.

Theoretical work on SGMs pursues a twofold goal: understanding *how* the approximate backward diffusion converges to the distribution of interest, and proving *that* it does. While it is clear that approximating the score function and using an approximate distribution both introduce errors in generation, it remains an open question under which conditions these approximations are sufficient to guarantee convergence to the true distribution. To establish such guarantees, existing works typically

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

rely on strong assumptions on the data distribution. For instance, [Bortoli, 2022] restricts to bounded support, [Gao et al., 2025] requires log-concavity, and [Conforti et al., 2025] assumes a finite moment of order eight. More broadly, all existing theoretical analyses of SGMs impose some form of strong assumption on the data distribution [Bruno and Sabanis, 2025, Silveri and Ocello, 2025, Bortoli et al., 2021, Strasman et al., 2025].

A more fundamental issue concerns the backward SDE itself. The founding paper on continuous SGMs [Song et al., 2021] relies on [Anderson, 1982] to introduce it, yet [Anderson, 1982] establishes this result only for distributions with bounded support. While other works [Bortoli et al., 2021] invoke alternative references [Haussmann and Pardoux, 1986], it remains unclear what minimal assumptions on the data distribution are required to rigorously justify the backward SDE, and therefore the diffusion framework itself.

Understanding for which distributions the diffusion framework can be rigorously introduced, and when convergence can be established, is thus a question of both theoretical and practical importance. On the practical side, the recent surge in high-quality image generation has prompted researchers to explore whether diffusion can be extended to heavy-tailed distributions. Several approaches have been proposed, including normalizing flows [Hickling and Prangle, 2025, McDonald et al., 2022, Laszkiewicz et al., 2022], variational autoencoders [Lafon et al., 2023], GANs [Allouche et al., 2022, Huster et al., 2021], and diffusion-based methods [Pandey et al., 2025, Yoon et al., 2022, Shariatian et al., 2025], yet none has fully solved the problem. Several works [Pandey et al., 2025, Shariatian et al., 2025] explicitly highlight the difficulties SGMs face in this regime, but it remains unclear whether these failures stem from fundamental limitations of the framework or from practical issues such as training instability, poor initialization, or SDE discretization.

In this paper, we show that early stopping and proper initialization together allow the diffusion framework to be applied, theoretically, to any distribution. The idea of improving generation through non-standard initialization has already been explored [Zheng et al., 2023, Zand et al., 2024, Frey et al., 2024, gil Lee et al., 2022, Shen et al., 2025, Scholz and Turner, 2026], but always with the goal of improving efficiency or generation quality. Our contribution is fundamentally different: we provide a theoretical guarantee that diffusion models can be applied to *any* distribution in the sense of KL convergence, without any assumption on the latter.

Precisely, this paper makes the following contributions.

- We prove that the backward SDE is well-posed for any data distribution, establishing that the diffusion framework is theoretically grounded in full generality.
- We prove that combining early stopping with a flexible initialization scheme that replaces the standard Gaussian prior is sufficient to guarantee convergence of the approximate diffusion to the target distribution without any assumption on the latter, generalizing [Conforti et al., 2025, Theorem 1] to this end.
- We validate our initialization-aware approach on standard benchmarks, demonstrating that a well-chosen initialization improves generation quality while reducing the number of required denoising steps.
- We take a first step toward modeling heavy-tailed distributions with standard diffusion models. We discuss what families of distributions are suitable choices for the initialization prior, and provide empirical evidence on toy datasets, synthetic climatic data, and real climatic datasets that proper initialization meaningfully improves over standard diffusion in this regime.

2 Generalized Variance Exploding diffusion

We discuss in this section the existence of the forward and backward diffusion equations in the context of Variance Exploding framework. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathrm{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ the space of square-integrable random variables on Ω. We denote by μ_{data} the unknown data distribution on $\mathbb{R}^d$, and the goal of SGMs is to generate new samples from μ_{data} given only an observed dataset.

2.1 Forward and Backward Process

Given $\overrightarrow{X}_0 \sim \mu_{\text{data}}$ the distribution of interest and the standard forward Variance Exploding (VE) equation [Song et al., 2021]

$$\mathrm{d}\overrightarrow{X}_t = g_t \, \mathrm{dB}_t \ . \tag{1}$$

where $g_t : \mathbb{R}^+ \to \mathbb{R}$ is assumed to be bounded and infinitely differentiable, we can state by applying Karatzas and Shreve [1991, Theorem 2.5] without any assumption on the data distribution that the forward solution $(\overrightarrow{X}_t)_{t\in[0,T]}$ is defined on all intervall for $T > 0$. Defining

$$\sigma_t^2 \coloneqq \int_0^t g_s \, \mathrm{d}s \ .$$

It can be stated that the solution

$$\overrightarrow{X}_t = \overrightarrow{X}_0 + \int_0^t g_s^2 \mathrm{d}s \overset{\mathcal{L}}{=} \overrightarrow{X}_0 + \sigma_t Z \ ,$$

Therefore as well known in the literature the solution simply results to be the convolution of the data distribution with a gaussian. In particular, for $t > 0$, $\overrightarrow{X}_t$ admits a density $\vec{p}_t(x)$ with respect to the Lebesgue measure $\lambda_{\mathbb{R}^d}$ and this density is highly regular. In particular the function $(t, x) \mapsto \vec{p}_t(x) \in C^\infty([\delta, T] \times \mathbb{R}^d)$, as shown in Lemma —- .

Concerning the backward equation the situation, might seem more complicated. The existence of the backward equation is not given for any distribution and any forward sde admitting an explicit solution. In fact the sufficient condition stated in reference articles [Anderson, 1982, Haussmann and Pardoux, 1986] require the data distribution or the density of the forward (when available) to respect some regularity conditions. Thankfully in the (VE) context where the forward solution is simply the convolution of the distributin with a gaussian, this doesn't represent a problem as stated in our first proposition.

Proposition 2.1. *Given μ_{data} a distribution and $\overrightarrow{X}_t, \vec{p}_t(x)$ the solutions of the* (1) *the backward equation*

$$\mathrm{d}\overleftarrow{X}_t = \bar{g}_t^2 \nabla \log \vec{p}_{T-t}(\overleftarrow{X}_t)\mathrm{d}t + \bar{g}_t \, \mathrm{d}\tilde{\mathrm{B}}_t \ , \tag{2}$$

where $\bar{g}_t \coloneqq g_{T-t}$ and $\tilde{\mathrm{B}}_t$ is an independent Brownian motion. For all initialization $\overleftarrow{X}_0$ and for all $\delta > 0$ (2) *admits a strong and unique solution $\overleftarrow{X} = \{\overleftarrow{X}_t\}_{[0,T-\delta]}$ on the interval $[0, T-\delta]$. In particular if $\overleftarrow{X}_0 \sim \vec{p}_T$, for all $\delta > 0$ it holds $\overleftarrow{X} \overset{\mathcal{L}}{=} \{\overrightarrow{X}\}_{[\delta,T]}$.*

The proposition that we presented establishes the existence of a backward strong solution for all

2.2 KL control on approximated process

In practice, μ_* is not available; instead, we only have access to an i.i.d. sample $\overrightarrow{X}_0^i \sim \mu_*$, $1 \le i \le n$. Consequently, the solution of Equation (2) that would enable sample generation is not directly accessible, since neither the score nor the terminal distribution $\vec{p}_T$ of the backward process are known. Furthermore, the backward SDE does not admit a closed-form solution, making it necessary to rely on discretization schemes.

For the purpose of theoretical analysis, following common practice, we may rely on either the Euler–Maruyama scheme Gao et al. [2025], Silveri and Ocello [2025] or the Exponential Euler–Maruyama scheme Conforti et al. [2025], Strasman et al. [2025]. Remarkably, in the VE setting these two schemes coincide, allowing for a unified treatment of the KL bound.

Score approximation and discretization. By replacing the real score function $\nabla \log \vec{p}_{T-t}$ using a trained neural network $s_\theta(T-t, \cdot)$, we introduce the discretized backward dynamics

$$\mathrm{d}X_t^\theta = \bar{g}_t^2 s_\theta(T - t_k, X_{t_k}^\theta) \, \mathrm{d}t + \bar{g}_t \, \mathrm{d}\tilde{\mathrm{B}}_t \ , \tag{3}$$

for $t \in [t_k, t_{k+1}]$, where $t_{k+1} = t_k + h_{k+1}$, with $t_0 = 0$ and $t_N = T - \delta$ which can be equivalently written, for $t \in [0, T-\delta]$, as

$$\mathrm{d}X_t^\theta = \sum_{k=0}^{N-1} \bar{g}_t^2 s_\theta(T - t_k, X_{t_k}^\theta) 1_{t\in[t_k,t_{k+1}]} \, \mathrm{d}t + \bar{g}_t \, \mathrm{d}\tilde{\mathrm{B}}_t \ . \tag{4}$$

This discretization scheme results in the following stochastic sampling dynamics:

$$X^{\theta}_{t_{k+1}} = X^{\theta}_{t_k} + (\sigma^2_{T-t_k} - \sigma^2_{T-t_{k+1}}) s_\theta(T - t_k, X^{\theta}_{t_k}) + \sqrt{\sigma^2_{T-t_k} - \sigma^2_{T-t_{k+1}}} Z \ , \tag{5}$$

for $k = 0, ..., N-1$, $Z \sim \mathcal{N}(0, \mathrm{I}_d)$. The only assumption we have to make to establish our result is a score approximation assumption :

- p^{θ}_0 and $\lambda_{\mathbb{R}^d}$ are mutually absolutely continuous.
- $\mathrm{D}_{\mathrm{KL}}\left(\vec{p}_t || p^{\theta}_0\right) < \infty$

H1 There exists $\varepsilon > 0$ such that for all $0 \leq k \leq N-1$,

$$\mathbb{E}\Big[\|\nabla \log \vec{p}_{T-t_k}(\overleftarrow{X}_{t_k}) - s_\theta(T - t_k, \overleftarrow{X}_{t_k})\|^2\Big] \leq \varepsilon \ .$$

We say that

$$\int_{\mathbb{R}^d} \log \frac{p_T}{p^0_\theta}(x) p_T(x) dx$$

is finite only in certain scenarios. Therefore : gaussian is not sufficient, the family of p_θ has at least to respect :

$$\sup_{\theta \in \theta} \int_{\mathbb{R}^d} \log \frac{p_T}{p^0_\theta}(x) p_T(x) dx < +\infty$$

Passando per l informazione di Fisher riesco a dimostrare che é sempre nulla.

3 Experiments

- We present experiments with normal datasets. We show that everything works well and we also saved some time. This is not new so this part will not be central.
- We present experiments with heavytail distribution. We show that it works pretty well, so that there is hope and ongoing work to do to correctly choose the flow p_θ which is exactly the point of research where we are stuck.
- We can include Nicolas and he could help me with doing the simulations on the heavytail datasets. As datasets i propose the synthetic dataset i created and the Comephere data which can be cleaned by me and Nicolas.

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